A FIELD NOTE ON SELF-IMPROVING AGENTS · NOVEMBER 2026

# When agents learn from failure, how do we keep them from learning the wrong lesson?

A controlled API-agent study of noisy feedback, grounded reflection, and the cost of getting better.

20-task training pool | 40 held-out tasks | Trace-grounded rules | A0 · A0-raw · A3

**The core challenge:** a failed request is evidence that something went wrong. It is not automatically evidence that the agent's instructions need a new rule.

# Two papers point to the same gap

Agents can improve by turning experience into persistent prompts, memories, or playbooks. The hard part is deciding which experiences deserve to change what the agent does next.

MEASURE LEARNING, NOT JUST ABILITY

### Agent Plasticity

Singh and colleagues argue that an agent should be evaluated on how well it improves on future held-out tasks, how efficiently it learns, and where the improvement loop breaks down. A high final score and an efficient learner are not necessarily the same thing.

Read the paper · arXiv:2610.08902

CURATE THE LEARNING SIGNAL

### GEPA-Curator

Yu and colleagues focus on the possibility that a failure comes from the task, harness, tools, environment, or judge rather than the prompt. Their Curator selects which experiences should inform optimization and can retrieve a prior successful trajectory as evidence.

Read the paper · arXiv:2610.37651

THE PROJECT'S TARGET

## Noise can become a persistent instruction

In this repository, an agent works through a synthetic REST API. It may receive an error because it omitted a required header, sent a bad field, hit an impossible task, or encountered unreliable feedback. A Reflector that generalizes from the error without enough evidence can add a harmful playbook rule. That rule then affects later tasks.

**Our question: can trace evidence and task curation make the learning loop more reliable—and do any gains transfer to held-out tasks for a reasonable cost?**

This is an inspired, small-scale testbed, not a reproduction of either paper's benchmarks or headline results.

# Make the evidence visible before changing the agent

The system turns one episode into a reviewable update proposal, then checks the proposal before it becomes persistent behavior.

**01 · ACT**

**Run a task**

The actor calls tools in the mock REST API.

**02 · TRACE**

**Capture evidence**

Keep the failed request, error body, and later success if observed.

**03 · REFLECT**

**Propose a rule**

The Reflector explains what to change and why.

**04 · CHECK**

**Ground, then gate**

Validate evidence; test grounded candidates on the batch.

## Three arms isolate the changes

| Arm | What changes | What it tells us |
|---|---|---|
| **A0** | Grounding check plus batch gate. | Baseline for evidence-constrained reflection. |
| **A0-raw** | Same Reflector prompt and gate; no grounding validator. | Whether grounding helps beyond aggregate gating. |
| **A3** | Curator can defer tasks and supply relevant prior successes; grounding and gate remain. | Whether selecting experiences adds value. |

## Grounding and gate checks

A rule must cite a supplied request/error pair and name a supported header, parameter, or field. A proposed value needs trace support unless the error itself names a missing header. Explicit rules that forbid fields used successfully are rejected. Only grounded proposals reach the gate.

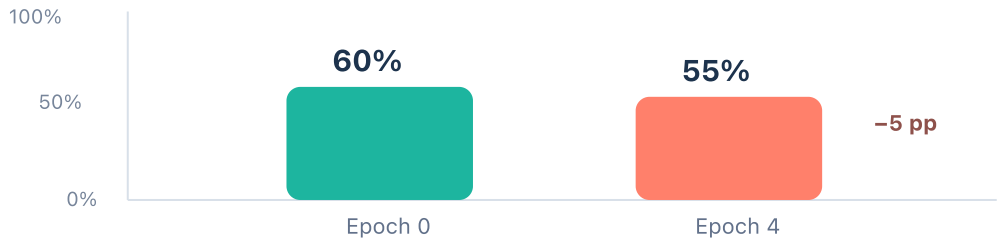
## Important limit

The gate compares aggregate training-batch scores. It can accept a candidate while one task regresses, and it cannot promise a held-out gain. Held-out evaluation is kept separate for that reason.

# A useful warning, not a verdict

The saved A0 checkpoint is incomplete. Its held-out score is lower at epoch 4 than at baseline; a newer attempt has now stopped in epoch 6 on repeated HTTP 429 errors.

**Held-out ID success · A0, seed 42** · same set of 40 tasks




**5 / 8**
epochs saved so far

**0**
rules in the saved epoch-5 playbook

**$0.111**
hypothetical tracked cost to epoch 5

## What the run log says

- Epoch-0 held-out: 24/40; epoch-4: 22/40.
- Training batches passed 7/10, 5/10, 5/10, 6/10, then 9/10 across epochs 1–5.
- At the last local log snapshot: 1,197 successful call records and 643 HTTP 429 retry events, followed by another reported 429 failure in epoch 6.

## How to read this

The saved checkpoint remains at epoch 5. The epoch-6 attempt exhausted three whole-episode retries and was not scored. This is one seed; the 5-point difference does not establish that A0 harms performance or that grounding caused the change.

Snapshot from local `seed-42-A0-full` results and call logs, read 11 Nov 2026; the epoch-6 error is from the subsequent console output. Cost is hypothetical reference pricing, not provider billing. Log counts can grow as a resumed run continues.

# One run can show where to look next

The current result points to two practical questions: can the run finish reliably under provider rate limits, and do grounded or curated updates improve held-out results across seeds?

### Completion plan

1. Resume A0 seed 42 to epoch 8 and evaluate at epoch 8.
2. Run the matched A0 seed 43, A0-raw seeds 42 and 43, and A3 seeds 42 and 43.
3. Compare held-out success at epochs 0, 4, and 8; include costs, rules, grounding/gate rejections, regressions, and 429 counts.
4. Report seed-level outcomes and uncertainty. Do not collapse an incomplete run or a two-seed result into a broad causal claim.

## Try the short run

Install the project dependencies, start local Phoenix, set `NVIDIA_API_KEY`, then run from the repository root:

```
python run_real_experiment.py --arm A0 --seed 42 --epochs 3 --eval-epochs 0,3 --run-tag demo
```

Use `--run-tag` for a fresh output folder; reuse the same tag to resume. Hosted runs make real model calls. The offline test suite does not.

THE TAKEAWAY SO FAR

**Reflection needs evidence, and evidence needs evaluation.** Grounding can constrain what a Reflector is allowed to generalize; curation can decide which failures deserve attention. Whether either approach improves learning efficiency is still an empirical question for the completed multi-seed comparison.

**Sources**
Singh et al., Agent Plasticity: Measuring Self-Improvement Through Experience.
Yu et al., GEPA-Curator: Curated Learning for Self Improving Agents.
Style reference: Arize, "Does Being Rude to AI Work?". This brief uses an original narrative and design.